

“CÁLCULO DE TAMAÑO”



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INTEGRANTES

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1.- Universidad

1.- Universidad			
Datos:		Fórmula:	
- $N = 12000$		$n = \frac{N z^2 p q}{E^2 (N-1) + z^2 p q}$	$n = \frac{12000 (1.96)^2 (0.5)(0.5)}{(0.05)^2 (12000 - 1) + (1.96)^2 (0.5)(0.5)}$
- $Z = 1.96$			$n = \frac{12000 (3.8416) (0.25)}{(0.0025)(11999) + (3.8416) (0.25)}$
- $p = 0.5$			
- $q = 0.5$			
- $E = 0.05$			$n = \frac{11524.8}{30.9579}$
		$n = 373$ estudiantes.	$n = 372.20 //$

2.- Hospital

2: Hospital

$$n = \frac{8500 (2.576)^2 (0.5)(0.5)}{(0.04)^2 (8500 - 1) + (2.576)^2 (0.5)(0.5)}$$

Datos:

$$n = \frac{8500 (6.6358) (0.25)}{(0.0016)(8499) + (6.6358)(0.25)}$$

$$- N = 8500$$

$$(0.0016)(8499) + (6.6358)(0.25)$$

$$n = 925 \text{ pacientes.}$$

$$- Z = 2.576$$

$$- p = 0.5$$

$$- q = 0.5$$

$$- E = 0.04$$

$$n = \frac{14701.07}{15.2573}$$

$$n = 924.08 //$$

3.- Empresa de Tecnología

Empresa de tecnología

Datos:

$$N = 20000$$

$$Z = 1.645$$

$$P = 0.5$$

$$q = 0.5$$

$$E = 0.03$$

$$N = \frac{20000(1.645)^2(0.5)(0.5)}{(0.03)^2(20000-1) + (1.645)^2(0.5)(0.5)}$$

$$N = \frac{20000(2.706)(0.25)}{(0.0009)(19999) + (2.706)(0.25)}$$

$$N = \frac{135.30}{18.679} = 724.65 \quad n = 725 \text{ clientes}$$

4.- Municipalidad

Municipalidad

Datos

$$N = 50000$$

$$Z = 1.96$$

$$P = 0.5$$

$$Q = 0.5$$

$$E = 0.02$$

$$N = \frac{50000 (1.96)^2 (0.5)(0.5)}{(0.02)^2 (50000 - 1) + (1.96)^2 (0.5)(0.5)}$$

$$N = \frac{50000 (3.8416) (0.25)}{(0.0004) (49999) + (3.8416) (0.25)}$$

$$N = \frac{48020}{20.9596} = 2291.06$$

$$N = 2291 \text{ ciudadanos}$$

5.- Salario promedio de empleados

Datos:

$$\bar{x} = \$2.800$$

$$\sigma = \$500$$

$$n = 100$$

$$Z_1 = 1.96$$

$$\frac{500}{\sqrt{100}} = \frac{500}{10} = 50$$

$$1.96 \times 50 = 98$$

$$IC = 2.800 \pm 98$$

$$2.800 - 98 = 2.702$$

$$2.800 + 98 = 2.898$$

$$IC = < 2.702 ; 2.898$$

6.- Tiempo de espera en Hospital

Datos:

$$\bar{x} = 42 \text{ min}$$

$$\sigma = 10 \text{ min}$$

$$n = 64$$

Nivel de confianza $\alpha = 99\%$

$$Z = 2.575$$

$$\frac{\sigma}{\sqrt{n}} = \frac{10}{8} = 1.25$$

$$2.576 \times 1.25 = 3.22$$

$$IC: 42 \pm 3.22$$

$$42 - 3.22 = 38.78$$

$$42 + 3.22 = 45.22$$

$$IC: < 38.78; 45.22$$

7.- Consumo de Electricidad

Ejercicio

Una compañía eléctrica estudia el consumo promedio mensual de hogares

$$IC : < 350 - 1.645 \times \frac{60}{\sqrt{81}} \quad ; \quad 350 + 1.645 \times \frac{60}{\sqrt{81}}$$

$$350 - 1.645 \times \frac{60}{9} \quad ; \quad 350 + 1.645 \times \frac{60}{9}$$

$$350 - 1.645 \times 6.7 \quad ; \quad 350 + 1.645 \times \frac{60}{9}$$

$$350 - 11.0215$$

$$350 + 11.0215$$

$$338.9785$$

$$361.0215$$

Datos: $\bar{x} = 350 \text{ kWh}$

$\sigma = 60 \text{ kWh}$

$n = 81$

Nivel de confianza = 90% - 1.645

8.- Rendimiento académico

Ejercicio

Una Universidad desea estimar el Promedio de calificaciones de sus estudiantes

$$\begin{aligned} IC &= \left[14.8 - 1.96 \frac{2.5}{\sqrt{49}} \right] ; \left[14.8 + 1.96 \frac{2.5}{\sqrt{49}} \right] \\ &= 14.8 - 1.96 \left(\frac{2.5}{7} \right) ; 14.8 + 1.96 \frac{2.5}{7} \\ &= 14.8 - 1.96 (0.36) ; 14.8 + 1.96 (0.36) \\ &= 14.8 - (0.7) ; 14.8 + (0.7) \\ &= 14.1 ; 15.5 \end{aligned}$$

Datos:

$$\begin{aligned} \bar{x} &= 14.8 \\ a &= 2.8 \\ n &= 49 \\ \text{nivel de confianza} &= 95\% - 1.96 \end{aligned}$$

✦ MUCHAS ✦
gracias